

PEMBUATAN AI CHATBOT STEP-BY-STEP

A. Buat fondasi chatbot:

1. Vector Database (Zilliz Cloud) <https://cloud.zilliz.com/>

Langkah: Create cluster yang free version dengan setting default, konfigurasi lebih lanjut pakai python di github.

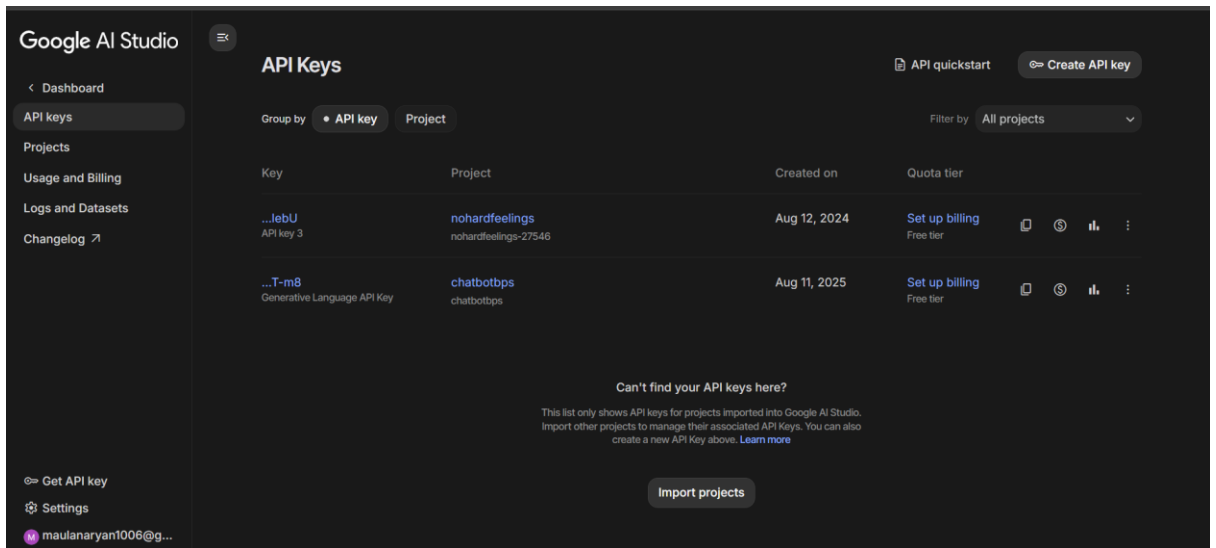
https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/create_collection%20zilliz%20vector%20db.py

The top screenshot shows the Zilliz Cloud interface. At the top, there is a banner for the 'Free Trial Ended' and a button to 'Add Payment Method'. The left sidebar contains navigation links: 'Default Project Enterprise', 'Clusters', 'Volumes', 'API Playground', 'Management', 'Backups', 'Migrations', 'Jobs', 'Collaborators', 'Project Alerts', and 'Network'. The main content area is titled 'Clusters' and features a table with the following columns: 'Name/ID', 'Region', 'Status', 'Deployment Option', 'Cluster Type', 'Query CU', 'Created At', and 'Actions'. A single cluster is listed: 'Free-01' in the 'gcp-us-west1' region, with a status of 'Resuming' and a deployment option of 'Free'. The bottom screenshot shows the 'Upgrade Deployment Option to Unlock More Features' dialog. It lists several benefits: 'Multi-replica and elastic scaling', 'Audit Logs', 'Private endpoint and VPC peering', 'SSO (SAML 2.0 based)', and 'And much more...'. It also shows connection details for the cluster: 'Cluster ID' (redacted), 'Region' (gcp-us-west1), 'Public Endpoint' (https://[redacted].serverless.gcp-us-west1.cloud.zilliz.com), 'Token' (redacted), and 'Private Endpoint' (Dedicated Only). The dialog also shows 'Compatible With Milvus 2.6.x', 'Created By 222212722@', and 'Created At Sep 9, 2025 1'.

2. AI API KEY (Gemini, Groq, Mistral) <https://aistudio.google.com/> , <https://console.groq.com/> , <https://admin.mistral.ai/>

Langkah: Create api key biasa dan catat key nya.

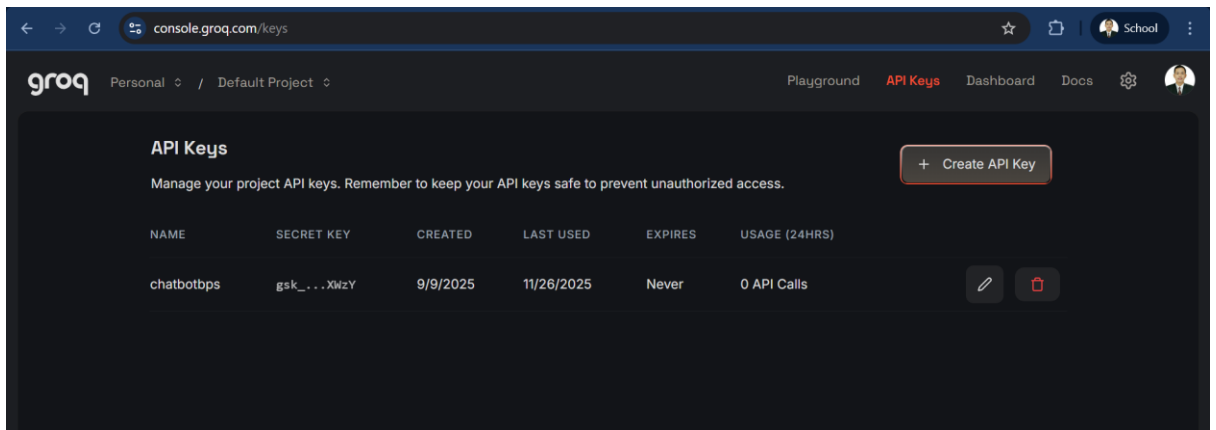
→ Gemini



The screenshot shows the Google AI Studio API Keys management interface. On the left is a sidebar with navigation links: Dashboard, API keys (selected), Projects, Usage and Billing, Logs and Datasets, and Changelog. The main area is titled 'API Keys' and includes a 'Create API key' button. A table lists existing API keys with columns for Key, Project, Created on, and Quota tier. Two keys are shown: one for 'nohardfeelings' created on Aug 12, 2024, and another for 'chatbotbps' created on Aug 11, 2025. Both are on the 'Free tier'. A message at the bottom states: 'Can't find your API keys here? This list only shows API keys for projects imported into Google AI Studio. Import other projects to manage their associated API Keys. You can also create a new API Key above. [Learn more](#)'.

Key	Project	Created on	Quota tier
...lebU API key 3	nohardfeelings nohardfeelings-27546	Aug 12, 2024	Set up billing Free tier
...T-m8 Generative Language API Key	chatbotbps chatbotbps	Aug 11, 2025	Set up billing Free tier

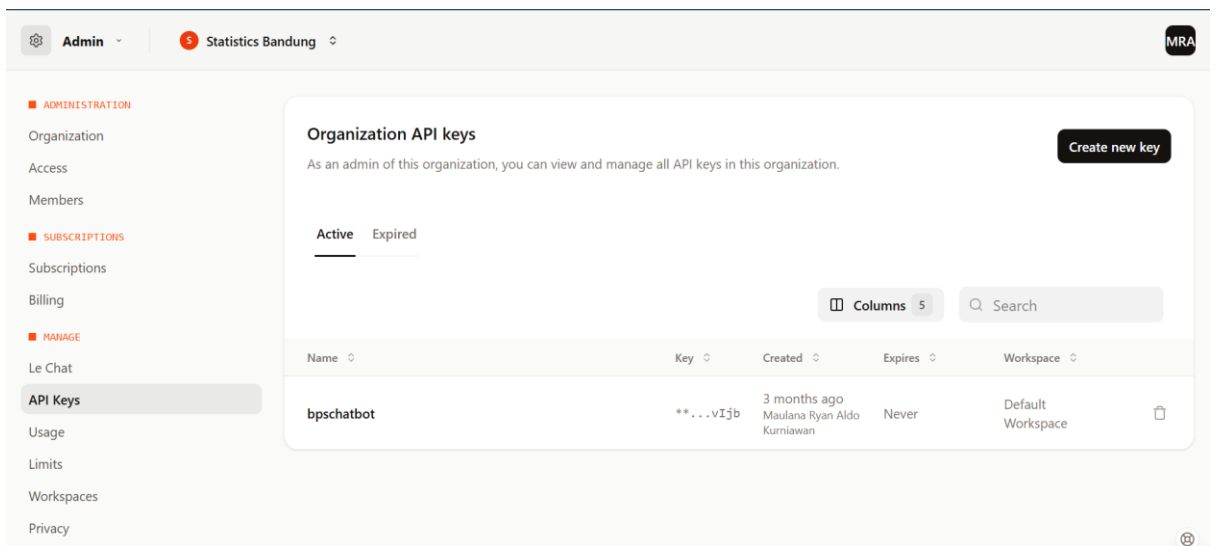
→ Groq



The screenshot shows the Groq console API Keys page. The browser address bar is 'console.groq.com/keys'. The page has a top navigation bar with 'Personal' and 'Default Project' dropdowns, and links for Playground, API Keys (selected), Dashboard, Docs, and a user profile. The main content area is titled 'API Keys' with a 'Create API Key' button. Below is a table with columns: NAME, SECRET KEY, CREATED, LAST USED, EXPIRES, and USAGE (24HRS). One key named 'chatbotbps' is listed with a secret key 'gs_k_...XWzY', created on 9/9/2025, last used on 11/26/2025, never expires, and has 0 API calls.

NAME	SECRET KEY	CREATED	LAST USED	EXPIRES	USAGE (24HRS)
chatbotbps	gs_k_...XWzY	9/9/2025	11/26/2025	Never	0 API Calls

→ Mistral AI

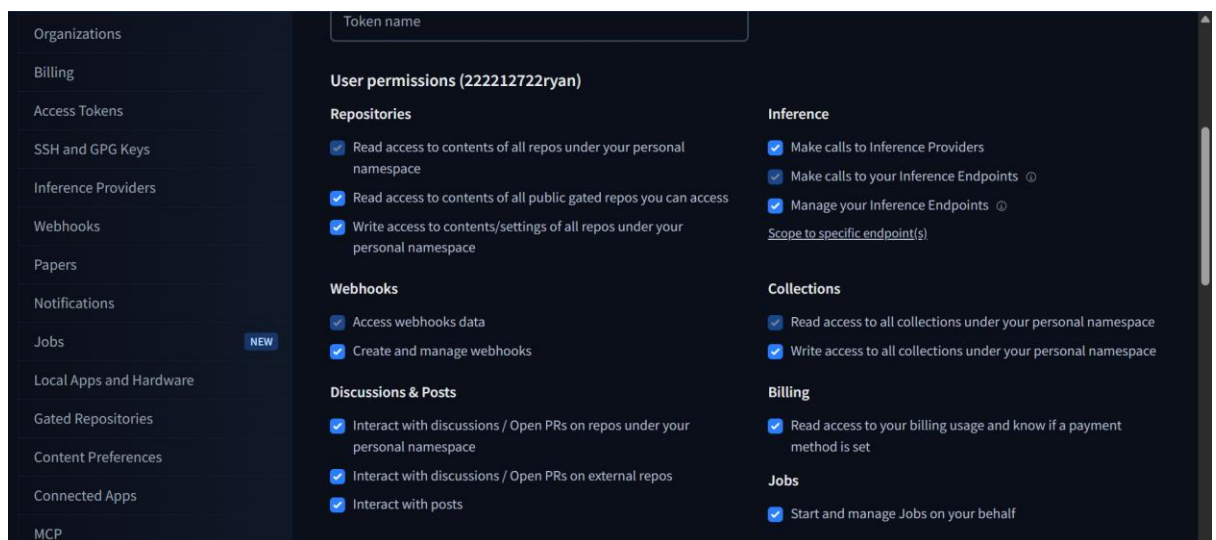
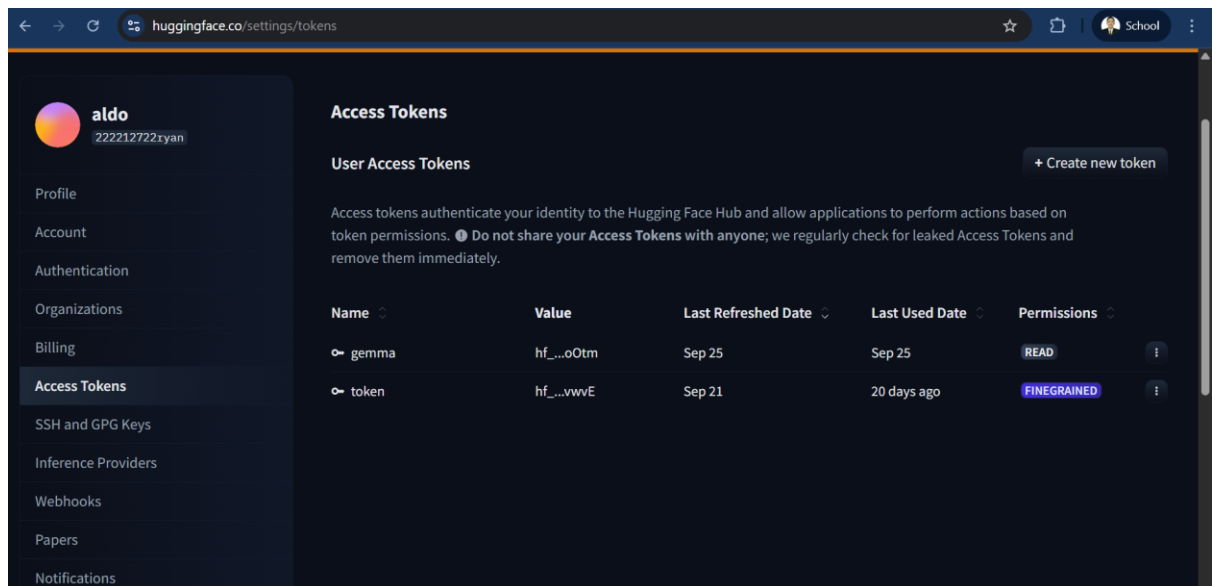


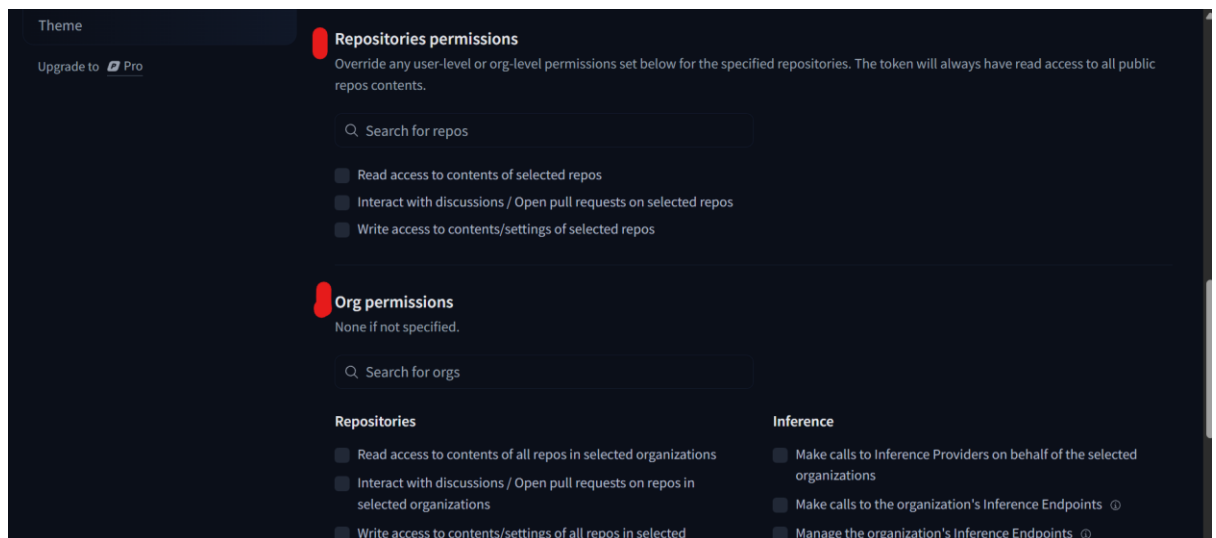
The screenshot shows the Mistral AI 'Organization API keys' management page. The top navigation bar includes 'Admin' and 'Statistics Bandung'. The left sidebar has sections for ADMINISTRATION (Organization, Access, Members), SUBSCRIPTIONS (Subscriptions, Billing), and MANAGE (Le Chat, API Keys (selected), Usage, Limits, Workspaces, Privacy). The main area is titled 'Organization API keys' with a 'Create new key' button. It features a toggle for 'Active' and 'Expired' keys. Below is a table with columns: Name, Key, Created, Expires, and Workspace. One key named 'bpschatbot' is listed with a key '***...vIjb', created '3 months ago' by 'Maulana Ryan Aldo Kurniawan', never expires, and is in the 'Default Workspace'.

Name	Key	Created	Expires	Workspace
bpschatbot	***...vIjb	3 months ago Maulana Ryan Aldo Kurniawan	Never	Default Workspace

3. Embedding model (HuggingFace) <https://huggingface.co/>

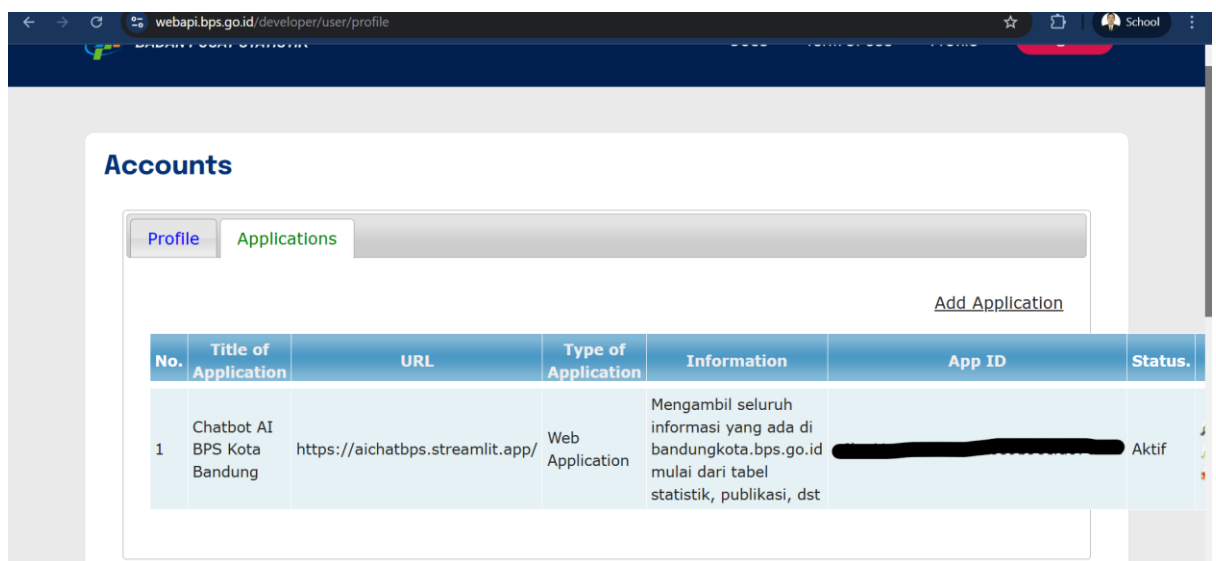
Langkah: create new token dan centang bagian user permission, untuk Repositories permissions dan Org permissions dikosongi saja.





4. WEBAPI BPS <https://webapi.bps.go.id/developer/>

Langkah: buat akun baru lalu add application.



B. Fetching dan cleaning data dari WEBAPI BPS

Karena fokusnya ke tabel saja, kita ambil data tabel statis, dinamis, dan simdasi.

Data versi jadi ada di **poin C**, berikut langkah-langkah fetching dan cleaning versi sederhana karena teknikalitas tinggi:

1. Ambil data dengan python di bawah.

<https://github.com/222212722->

[spec/chatbot_bps/blob/main/cara%20pengisian%20database/fetch_table](https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/fetch_table)

misal output dinamai: bps_data.json

2. Setelah punya data (json) berisi variabel penting dari tabel, kita olah karena json belum punya variabel pelengkap seperti link, primary key (unique), dll. Di sini salah satu teknik yang dipakai adalah slug (membuat variabel link berdasarkan title, id, tabelsource).

2.1 Variabel semula

Variabel	Keterangan
id	ID tabel (encoded dari API BPS)
title	Judul tabel
oldest_period	Tahun data paling lama
latest_period	Tahun data paling baru
id_subject	ID subjek statistik
subject	Nama subjek
id_subcat	ID subkategori
subcat	Nama subkategori
tablesource	Sumber tabel
last_update	Waktu update terakhir

2.2 variabel yang dipertahankan

Variabel	Asal	Catatan
id	dari data awal	Nilai berbeda karena konteks tabel berbeda
title	dari data awal	Biasanya dibersihkan (hapus prefix [Administrasi], (Orang))

Variabel	Asal	Catatan
id_subcat	dari data awal	Sama
subcat	dari data awal	Sama
tablesource	dari data awal	Bisa berubah nilai
last_update	dari data awal	Format disederhanakan (tanggal saja)

2.3 variabel baru

Variabel baru	Dibentuk dari
unique	Kombinasi id_table + suffix (static)
id_table	Ekstraksi / decode dari id
years	Dari latest_period atau range tahun
link	Slug dari title + id + tablesource
page_content	Template berbasis title
embedding	Hasil embedding dari page_content

Output: bps_data.json berubah jadi alltables_ready.json

Pakai python untuk eksekusinya, salah satu contohnya ada di link berikut:

https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/slug_title

3. Setelah semua variabel siap, lakukan embedding alias indexing dengan google colab.

https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/embed_gemma

output: alltables_ready.json berubah jadi alltables_embedded_gemma.json

C. Memasukkan Data yang telah diolah ke database

1. Isi database dengan data yang telah di embed (alltables_embedded_gemma.json)

Data tabel versi terbaru yang sudah di embed
<https://drive.google.com/file/d/1OimABpsUWuFgHh7R45KbPpAUgyLxkJzB/view?usp=sharing>

2. Eksekusi load collection dengan Python:

[https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/load_collection%20\(zilliz%20vector%20db\).py](https://github.com/222212722-spec/chatbot_bps/blob/main/cara%20pengisian%20database/load_collection%20(zilliz%20vector%20db).py)

Dengan demikian, database vector terisi dan chatbot bisa dijalankan.